

Literature status: the finite-fourth-moment Kolmogorov bound asked for in arXiv:1707.01889

Answer identified in arXiv:2607.28742

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Status: answered in later literature. Remark 1.4(b)–(c) of arXiv:1707.01889 asks for a Kolmogorov fourth-moment bound of the same square-root order as its Wasserstein bound, under only a finite fourth moment. Theorem 1.9 of arXiv:2607.28742 proves precisely

$$d_K(F, N) \leq 15.6 \sqrt{\mathbb{E}[F^4] - 3}$$

for every variance-one random variable in a fixed Poisson chaos with finite fourth moment.

1 Question in the source

Let F belong to the q th Poisson Wiener chaos, let $\sigma^2 = \mathbb{E}[F^2] > 0$, and assume only $\mathbb{E}[F^4] < \infty$. Theorem 1.2 of Döbler, Vidotto, and Zheng [1] gives the Wasserstein estimate

$$d_W(F, \sigma N) \leq C_{q,\sigma} \sqrt{\mathbb{E}[F^4] - 3\sigma^4}.$$

On printed page 4, Remark 1.4(b) observes that the generic comparison between Kolmogorov and Wasserstein distances loses a square root and yields only a fourth-root estimate. Remark 1.4(c) then asks whether one can obtain a Kolmogorov bound with the same square-root accuracy. At that time such a bound was known only under an additional local regularity assumption.

The mathematically concrete issue is therefore whether the extra regularity can be removed while retaining a bound proportional to the square root of the fourth cumulant.

2 Exact later answer

Zheng [2] states this problem as the motivation of the paper on its first page. Theorem 1.9 (printed page 5) says: if $q \geq 1$ and F lies in the q th Poisson chaos, $F \in L^4$, and $\mathbb{E}[F^2] = 1$, then

$$d_K(F, N) \leq 15.6 \sqrt{\mathbb{E}[F^4] - 3}.$$

This has exactly the requested exponent and assumes no regularity beyond the finite fourth moment. The paragraph following the theorem explicitly says that it removes Assumptions A and $\mathbf{A}^{\text{loc}}$ from the earlier Kolmogorov estimate of Döbler and Peccati.

The mechanism is a fixed-chaos martingale core. Theorem 1.7 constructs conditional expectations F_n which simultaneously remain in the original Poisson chaos, have bounded step kernels with finite-measure support, and converge to F in L^4 . The regular Kolmogorov estimate applies to F_n ; fourth cumulants converge, and lower semicontinuity of Kolmogorov distance (Lemma 2.2) passes the

estimate to F . Thus the later theorem supplies the missing closure argument rather than imposing the old local assumption.

For a non-unit variance σ^2 , applying the theorem to F/σ gives the equivalent scaled statement

$$d_K(F, \sigma N) \leq \frac{15.6}{\sigma^2} \sqrt{\mathbb{E}[F^4] - 3\sigma^4}.$$

3 Scope

This is a direct answer to the Poisson-chaos fourth-moment problem motivating Remark 1.4: the desired Kolmogorov accuracy holds under the weakest stated moment assumption. The 2026 proof is not advertised as a general black-box Kolmogorov theorem for every exchangeable-pair construction; it combines a martingale-core approximation with the established regular Poisson-chaos bound. Hence the concrete bound is settled, while the broadest possible methodological reading of “via exchangeable pairs” is not claimed here.

References

- [1] C. Döbler, A. Vidotto, and G. Zheng, *Fourth moment theorems on the Poisson space in any dimension*, arXiv:1707.01889; Electron. J. Probab. 23 (2018), Paper 36.
- [2] G. Zheng, *A Kolmogorov fourth-moment bound on Poisson chaos via a martingale core*, arXiv:2607.28742v1, 2026.